

Restoration of Motion-Blurred, High-Resolution Mars Express SRC Images of Phobos

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Highlights

What are the main findings?

- Developed a fully automated, reproducible pipeline for restoring motion-blurred Mars Express SRC images using geometry-derived PSFs and Wiener deconvolution.
- Demonstrated consistent recovery of sub-meter surface features on Phobos across 14 images from multiple orbits, including severe motion-smear cases.

What is the implication of the main finding?

- Enables enhanced scientific analysis of planetary surfaces from archival data without star-field calibration or manual kernel tuning.
- Provides a framework readily adaptable to other planetary missions affected by motion-induced blur, supporting future high-resolution mapping.

Abstract

We present an automated and fully reproducible pipeline for restoring motion-smeared Mars Express SRC images of Phobos. A one-dimensional motion point spread function (PSF) is derived directly from SPICE geometry and microsecond-precision exposure timing, and Wiener deconvolution (SNR = 16 dB) is applied to recover image sharpness. Tested on 14 images from 4 orbits spanning slant distances of 52–292 km, exposures of 14–20 milliseconds, sampling of 0.47–2.7 m/pixel, and PSF lengths of 11–119 pixels, the method achieves up to 31.7 dB PSNR, 0.78 SSIM, and positive sharpness gains across all cases. The restored images reveal sub-meter surface features previously obscured by motion blur, with residual energy reduced relative to the acquisition model. The workflow relies solely on open data and open-source tools (ISIS, ALE/SpiceyPy, OpenCV), requires no star-field calibration, and generalizes to other motion-degraded planetary datasets, providing a fully transparent and reproducible solution for high-resolution planetary imaging.

Keywords: Phobos; Mars Express; motion blur; image restoration; Wiener deconvolution; point spread function; spacecraft imagery; Super-Resolution Channel



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1. Introduction

Motion-induced blur (“motion smear”) degrades spacecraft imagery when the relative motion between the sensor and target is substantial, hindering detailed surface characterization. Existing correction methods often rely on star-field calibration, iterative kernel tuning, or manual parameter adjustment, which limit reproducibility and applicability across

diverse imaging conditions. Our approach directly derives the blur kernel from spacecraft geometry, providing a fully automated and calibration-free solution that addresses these gaps.

Notable cases are Mars Express's Super-Resolution Channel (SRC) images of both Phobos [1] and Deimos [2]. Although SRC can achieve sub-meter sampling and may offer the highest resolution images currently available for Phobos, residual "pepper" noise (see Supplementary Information and Figure S1 for details) and severe smear have limited its scientific exploitation. Previous SRC image restorations often relied on star-field calibration or iterative kernel estimation with limited reproducibility. Our SPICE-driven Wiener deconvolution pipeline is the first to automatically derive the PSF solely from spacecraft geometry and apply a one-shot, non-iterative restoration with built-in noise regularization, enabling full reproducibility and operational efficiency across diverse imaging conditions.

2. Materials and Methods

2.1. Dataset Selection and Pre-Processing

As the spacecraft approaches a target body, the resulting images would become both more blurred and higher in resolution. Thus, using the European Space Agency (ESA) Planetary Science Archive (PSA) web interface, we first listed Mars Express High Resolution Stereo Camera (HRSC) [3] Super-Resolution Channel (SRC) level 3 data products by setting slant distance of less than 300 km to Phobos. The <300 km slant-distance cutoff ensures sufficient spatial resolution (<3 m/pixel), while excessive motion blur at closer ranges reduces image quality. We excluded images with slant distances >300 km because their sampling exceeds ~3 m/pixel, at which point higher-resolution images already exist (e.g., the Mars Global Surveyor's Mars Orbiter Camera images). Moreover, at such distances, the spacecraft motion during exposure produces only very short PSFs, meaning that SRC images contain little visible blur and benefit less from deblurring.

We browsed all publicly accessible datasets acquired within observation dates since 2003 to the latest release (as of June 2025). After that, we manually selected SRC image IDs whose boresight intersects visible (not dark) surfaces of Phobos by visual inspection. This finally achieved a list of fourteen highest-resolution (better than a few meters per pixel) SRC images which were observed during orbits 5851 (23 July 2008), orbit 7926 (10 March 2010), orbit 8974 (9 January 2011), and orbit 14776 (26 August 2015) (see Supplementary Table S1 for details), three of which are consistent with those listed by Witasse et al. [4].

Based on that, we acquired Planetary Data System (PDS) IMG datasets of HRSC radiometric Reduced Data Record (RDR) Extension series (EXT2, EXT3, EXT5, and EXT7) Version 4.0 from the PDS Geosciences Node server (<https://pds-geosciences.wustl.edu/mex>, accessed on 18 September 2025). Also, we downloaded the MEx-related Spacecraft, Planet, Instrument, C-matrix, Events (SPICE) kernels [5] by running the Integrated Software for Imagers and Spectrometers (ISIS) version 8.3.0 developed by United States Geological Survey (USGS) [6], as well as an extra SPK kernel, MEX_STRUCT_V01.BSP, from the SPICE datasets in the ESA PSA server (MEX-E-M-SPICE-6-V2.0) [7]. Adding MEX_STRUCT_V01.BSP refines spacecraft attitude and structural geometry, improving alignment accuracy for PSF derivation.

We first converted their PDS IMG files to ISIS cubes and attached MEx SPICE kernels to the cubes by running both "hrsc2isis" and "spiceinit" commands of ISIS 8.3.0 (Figure 1). To obtain better shift in surface-intersection points to be required in the following image restoration, during "spiceinit", we add the latest, finest (~18 m mean spacing) shape model [8] as a DSK kernel that was converted from phobos_g_018m_spc_0000n00000_v002.obj (available at <https://sbmt.jhuapl.edu/shared-files/>, accessed on 18 September 2025) by using the Navigation and Ancillary Information Facility (NAIF) SPICE utility program

MKDSK (https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/ug/mkdsk.html, accessed on 18 September 2025). Prior to the following frequency-domain work, we denoise peppers/LRS (“Low Representation Saturation”) pixels in each cube (see Section S1 of the Supplementary Information and Supplementary Figures S1 and S2 for further details on peppers and the impact of remaining peppers after image restoration).

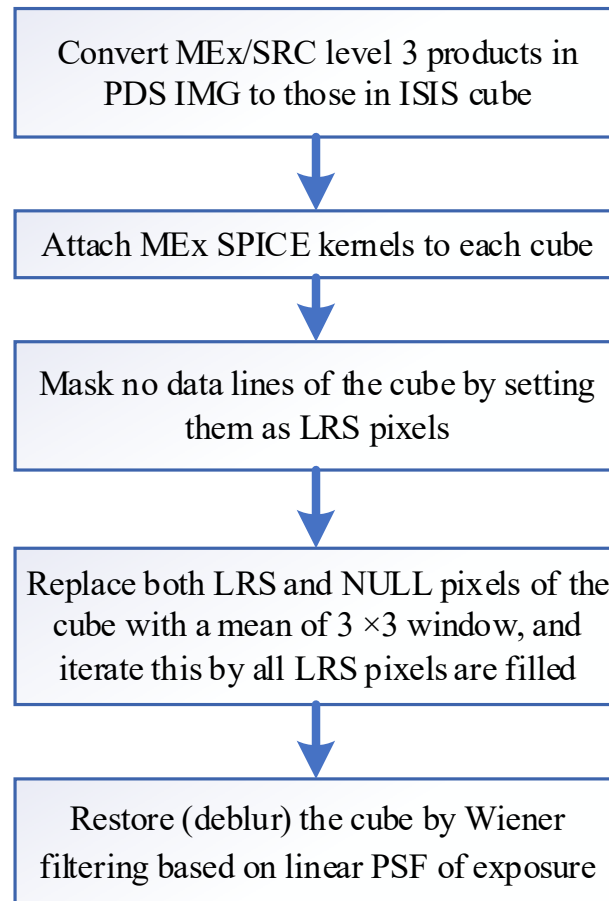


Figure 1. Workflow of the whole pipeline of our SRC image deblurring processing.

2.2. PSF Estimation

The motion-induced blur can be modeled as a one-dimensional linear point spread function (PSF). Each PSF can be parameterized by the spacecraft’s instantaneous ground-track velocity and the detector exposure time that are accessible by the USGS Abstraction Layer for Ephemerides (ALE) library [9]. These parameters are effectively determined by the spacecraft’s positions in body-fixed coordinates at the start and end of the exposure. Therefore, the resulting PSF’s length and its orientation (along the projected motion vector over Phobos) can be derived directly from spacecraft telemetry. See the Supplementary Information’s Section S2 (including Supplementary Figure S3) for schematic illustrations of the observation geometries and derivation of a PSF as a linear shift of surface-intersection points on the SRC image plane during exposure.

The obtained PSF geometries are summarized in Figure 2 and Supplementary Table S3. By using longer exposures—values matching the SRC observation durations—our pipeline extracted extended PSFs, which is consistent with the initial expectation of this study. Most of the vertical components of the shifts are close to zero relative to horizontal components (i.e., each shift is in parallel with its flight direction), so that the resulting PSF can be approximated to be a horizontally flat line (See Figure 2 or a typical case).

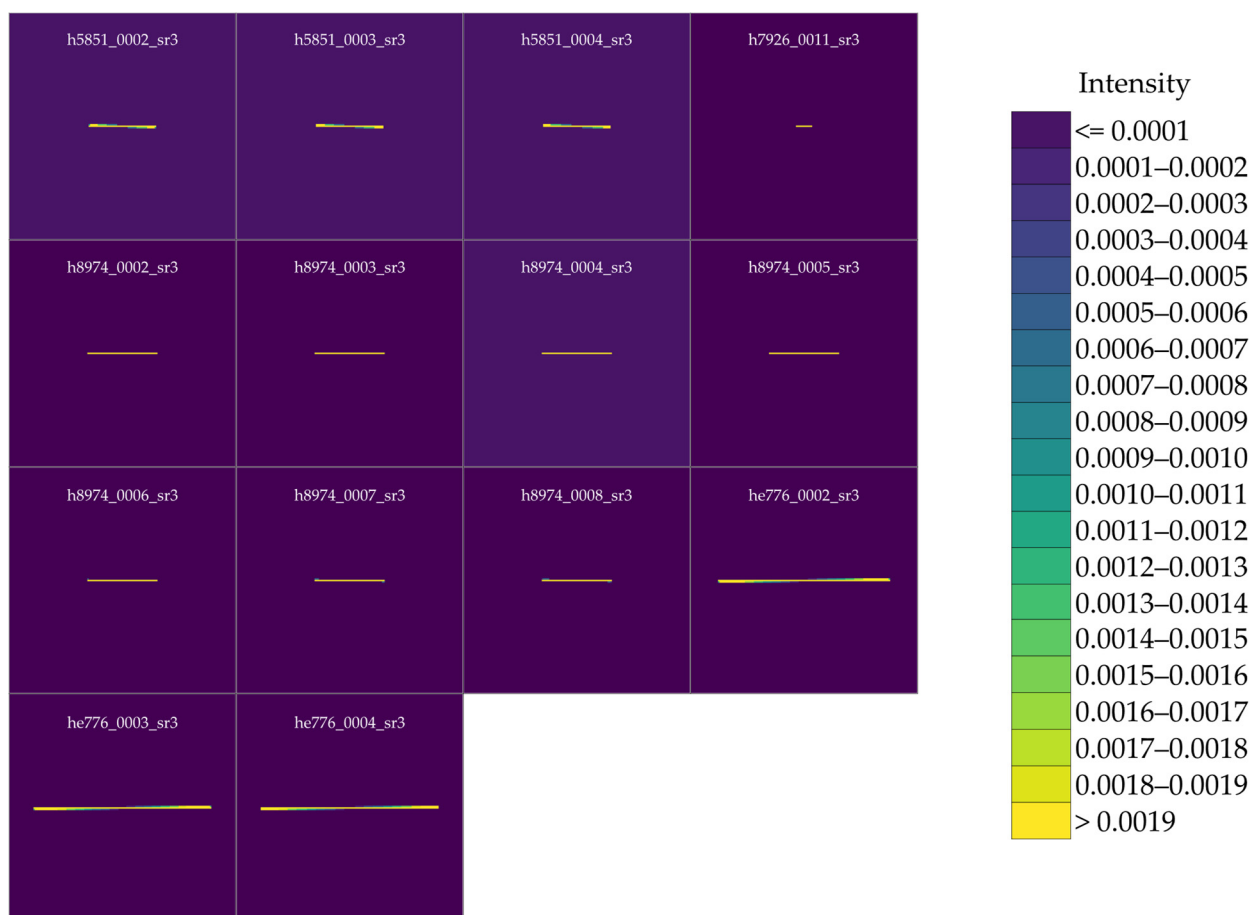


Figure 2. Results of spatial-domain intensity of phase-shifted PSFs calculated for each of the selected SRC images. The intensity per pixel is color-coded by intervals defined by the color bar on the right. Mostly, the highest intensity pixels lie horizontally at the center of each PSF, suggesting the along-track shift of the surface-intersection points in the SRC boresight vector during exposure. All pixels comprising each of the PSFs are normalized so that its total energy = 1. Each window size is 150×150 pixels. See Table S3 for their lengths in pixels.

2.3. Wiener Deconvolution

Wiener algorithm [10] computes a single “optimal” filter $H_W(u, v)$ that, when applied to the blurred image’s Fourier transform $G(u, v)$, yields an estimate of the true image’s transform $F(u, v)$:

$$\hat{F}(u, v) = H_W(u, v)G(u, v) \quad (1)$$

$$H_W(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + \frac{S_N(u, v)}{S_F(u, v)}} = \frac{H^*(u, v)}{|H(u, v)|^2 + \frac{1}{SNR}} \quad (2)$$

where (u, v) are the frequency coordinates corresponding to the spatial coordinates (x, y) , $H(u, v)$ (and hence $H_W(u, v)$) is the transfer function (Fourier-domain PSF), H^* is complex conjugate of H , $S_N(u, v)$ and $S_F(u, v)$ are the noise and signal power spectra, respectively (that is, $S_N(u, v)/S_F(u, v)$ is the noise-to-signal power ratio), signal-to-noise ratio (SNR)—a dimensionless quantity—is given by $SNR = S_F(u, v)/S_N(u, v)$, and $\hat{F}(u, v)$ is the estimate of the true image’s Fourier spectrum (see Section S3 of the Supplementary Information and Supplementary Table S4 for further discussion of the comparison with Lucy–Richardson deconvolution).

We used Wiener deconvolution with the linear PSFs aforementioned and an empirically determined $SNR = 16$ dB (i.e., $10^{(16/10)} \cong 39.8$ in linear scale), which works well regardless of a variety of SRC observation conditions. To mitigate serious ringing arti-

facts of output boundaries after deconvolution [11], we also used edge tapering algorithm (Gaussian-weighted taper at edges) with a width of 81 pixels on both right and left sides (and no tapering at top and bottom sides) of each input image beforehand (see Supplementary Figure S4 for further details on the edge-tapering effect). Although MATLAB offers built-in routines for Wiener deconvolution and edge tapering (e.g., `deconvwnr` and `edgetaper`, available since before R2006a), we chose to avoid reliance on proprietary software by reimplementing these methods in OpenCV. Specifically, we extended OpenCV's deconvolution sample (<https://github.com/opencv/opencv/blob/master/samples/python/deconvolution.py>, accessed on 18 September 2025) to include edge tapering, yielding a fully open-source solution.

3. Results

We successfully restored motion-blurred SRC images of the four MEx orbits, significantly reducing blur and enhancing geological feature visibility. Our pipeline establishes a robust and objective framework for future restoration and analysis of motion-blurred planetary imagery.

The selected fourteen images span four distinct orbits with varying spatial resolutions (0.47–2.7 m/pixel), exposure durations (~14–20 milliseconds), and ground-track velocities. This diversity ensures that the dataset captures a representative range of motion-blurred conditions present in the entire Mars Express SRC archive, from minimal to severe smear scenarios.

3.1. Restoration Results

3.1.1. Orbit 5851

Within the four selected MEx orbits (and most of all MEx orbits), the orbit 5851 has observed SRC images with the second highest resolutions (~0.86 m/pixel; Supplementary Table S1) and the shortest exposure duration (~14 milliseconds; Supplementary Table S2), which caused severe motion blurs with lengths of ~44 pixels (Supplementary Table S3) in resulting images. However, unclear surface features, such as impact craters and portions of lineaments, have appeared unequivocally after restoration, as shown in Figure 3. Noisy artifacts in images are still visible, but ringing artifacts at edges are minimized in restored images. As for small-scale craters, a diversity of their depths, rims, floors, and degradational states is also identifiable (e.g., a fresh, rayed crater on the top-right part of Figure 3b), which is consistent with morphologies of sub-kilometer-scale craters on Phobos [12] (see Supplementary Figure S5 for a detailed geological interpretation). Based on the above inspection, we conclude that our restoration pipeline restored the SRC images in orbit 5851 to a satisfactory level of quality.

3.1.2. Orbit 7926

The SRC image of MEx orbit 7926 has the lowest resolution and the longest slant distance (~2.7 m/pixel and ~292 km; Supplementary Table S1) among all the SRC images selected in this study. With an exposure duration as short as that of orbit 5851, the shortest PSF only ~11 pixels in length (Supplementary Table S3) caused a slight motion smear to the original image, which is consistent with its appearance in Figure 4a. After our restoration process, we obtained the image that exhibits the well-defined craters and grooves on Phobos (see Supplementary Figure S5 for a detailed geological interpretation). Although the imaging distance to the surface would be proximal on the left side and distal to the right side of the figure, the restoration appears successful on both sides. A few noises are identified on the resulting image; however, the edge tapering effect has not seriously

reduced the original image's quality. As a result, we determined that the SRC image of orbit 7926 was successfully restored.

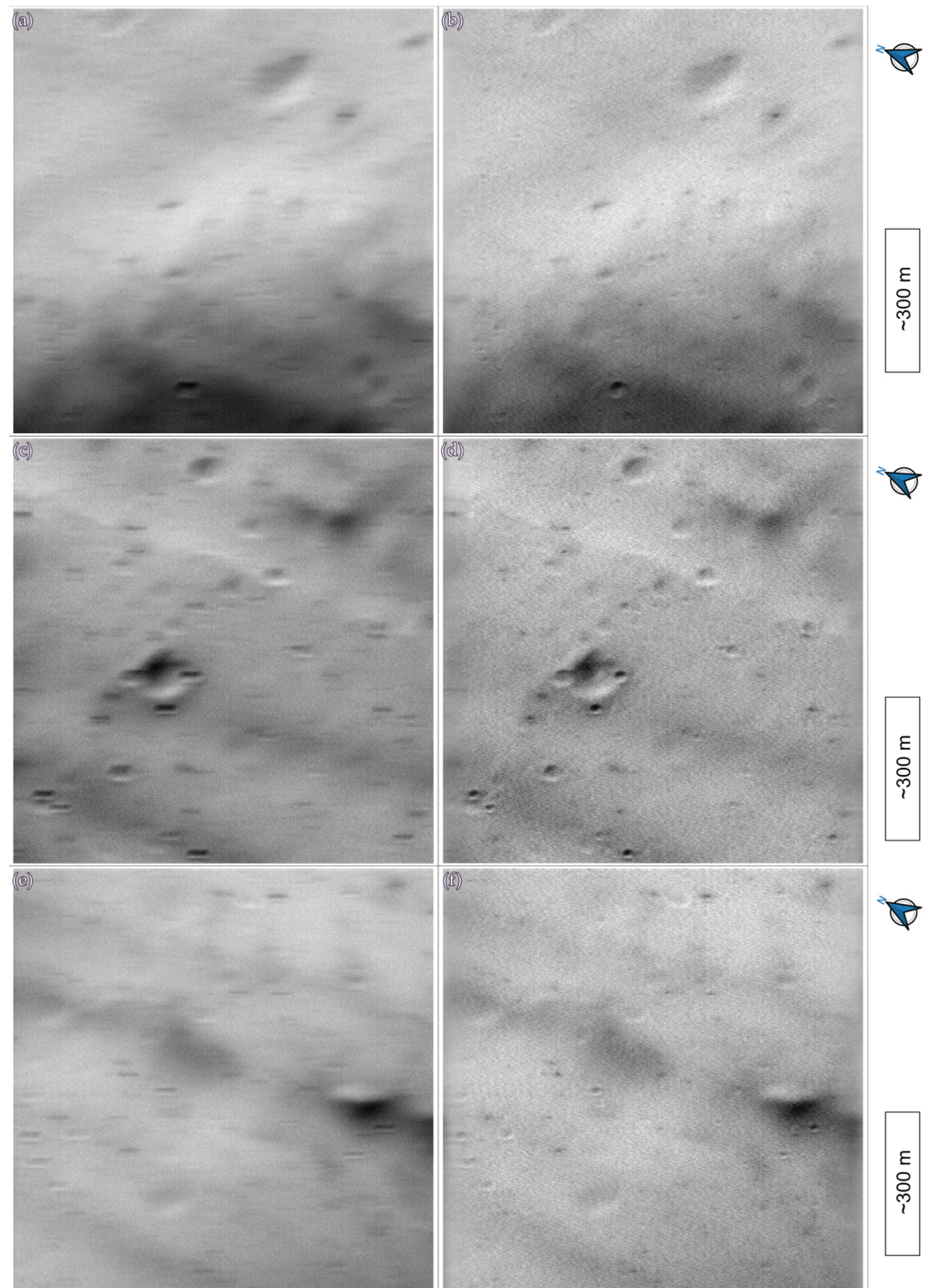


Figure 3. The result of deconvolution of three SRC images taken in MEx orbit 5851. (a,c,e) denoised images before restoration and (b,d,f) deblurred images after restoration. Images are ordered by SRC image ID (Supplementary Table S1). Scale bars and north arrows provide approximate spatial reference only; uncertainties in Phobos's shape model and spacecraft pointing/position may introduce errors of tens to hundreds of meters. Key surface features discussed in the text are left unannotated here to maintain image visibility.



Figure 4. The result of deconvolution of one SRC image taken in MEx orbit 7926. (a) a denoised image before restoration and (b) a deblurred image after restoration. Scale bars and north arrows provide approximate spatial reference only; uncertainties in Phobos's shape model and spacecraft pointing/position may introduce errors of tens to hundreds of meters. Key surface features discussed in the text are left unannotated here to maintain image visibility.

3.1.3. Orbit 8974

Seven images acquired in the MEx orbit 8974 have similar resolutions and slant distances to those of the orbit 5851 and a slightly longer exposure duration (~ 16.1 milliseconds; Supplementary Table S2) with their PSF lengths of ~ 46 pixels (Supplementary Table S3), which cover the southern hemisphere of anti-Mars surface of Phobos (Supplementary Table S1). As shown in Figure 5, all SRC images captured a wide range of crater sizes and morphologies, and some of them observed lineaments (Figure 5a) and slopes (Figure 5g), which would be significant for analyses in surface irregularity (e.g., [13]). After the deconvolution, the resultant images show well-defined outlines of surface features much better than those of the original images. Furthermore, we can identify several-pixel-scale positive relief features (most likely boulders; Figure 5d,f) that were difficult to identify from the original blurred images (see Supplementary Figure S5 for a detailed geological interpretation). Regardless of diverse landscapes, the deconvolved images show few noises and edge-tapered pixels at negligible levels. We express that deblurring the orbit 8974 images was achieved successfully.

3.1.4. Orbit 14776

Among the selected (and probably all available) MEx SRC images, those acquired in the MEx orbit 14776 are the worst motion-blurred, due to exposures of ~ 20 milliseconds (Supplementary Table S2), the closest distance (~ 52 km), and the highest resolution (~ 0.47 m/pixel; Supplementary Table S1), which is the hardest to recover by deconvolution. However, as shown in Figure 6, we performed restoration and acquired the images showing identifiable surface features, such as small impact craters and boulder-like positive reliefs (e.g., small knobs on both top and right sides of Figure 6c; see Supplementary Figure S5 for a detailed geological interpretation). Even though there are noticeable noises and edge-tapering effects left in the resulting images, given the initial bad imaging conditions, we conclude our efforts made them as restored as possible, from a theoretical viewpoint.

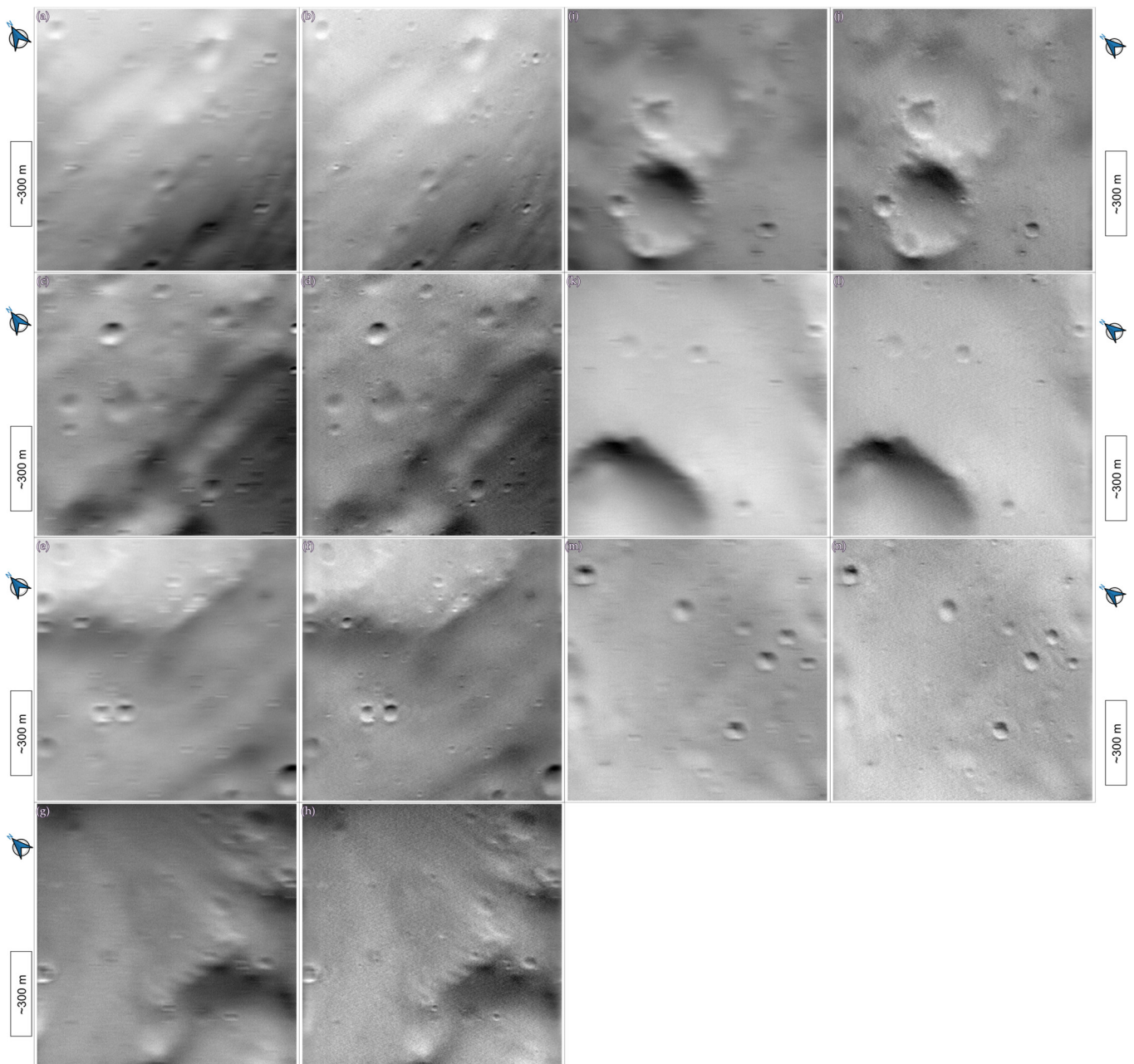


Figure 5. The result of deconvolution of seven SRC images taken in MEx orbit 8974. (a,c,e,g,i,k,m) denoised images before restoration and (b,d,f,h,j,l,n) deblurred images after restoration. Images are ordered by SRC image ID (Supplementary Table S1). Scale bars and north arrows provide approximate spatial reference only; uncertainties in Phobos’s shape model and spacecraft pointing/position may introduce errors of tens to hundreds of meters. Key surface features discussed in the text are left unannotated here to maintain image visibility.

3.2. Quantitative Validation

We quantitatively evaluated restoration quality using four complementary metrics: (i) the forward-model residual $R = G - (h * \hat{F})$, reporting its root mean square (RMS) over the untapered image region; (ii) a sharpness gain, defined as the ratio of the variance of the Laplacian before and after restoration $Vol_{gain} = Var[\nabla^2 \hat{F}] / Var[\nabla^2 G]$; and (iii) data consistency via the Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) between the re-blurred image $h * \hat{F}$ and the denoised input G , i.e., $PSNR(G, h * \hat{F})$ and $SSIM(G, h * \hat{F})$.

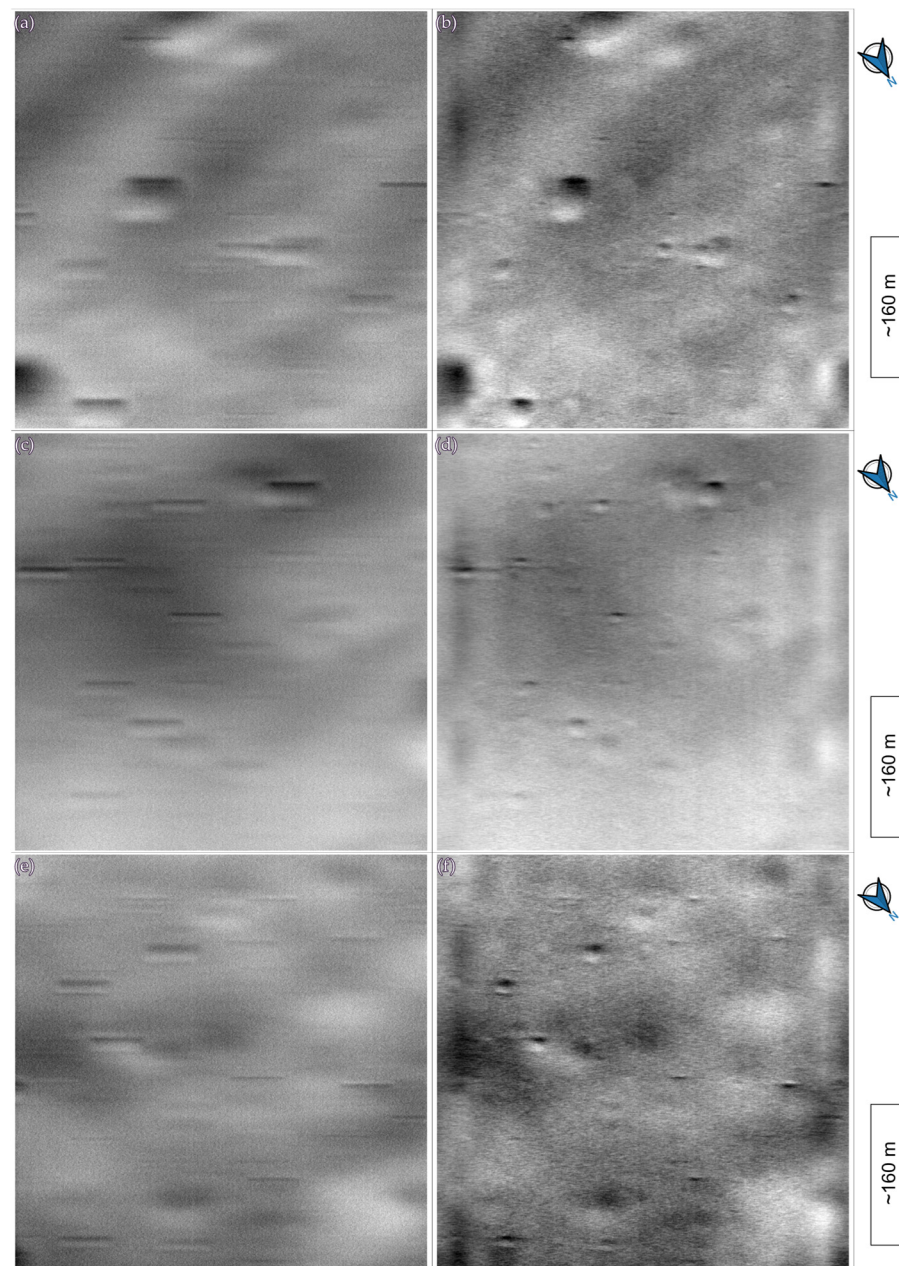


Figure 6. The result of deconvolution of three SRC images taken in MEx orbit 14776. (a,c,e) denoised images before restoration and (b,d,f) deblurred images after restoration. Images are ordered by SRC image ID (Supplementary Table S1). Scale bars and north arrows provide approximate spatial reference only; uncertainties in Phobos’s shape model and spacecraft pointing/position may introduce errors of tens to hundreds of meters. Key surface features discussed in the text are left unannotated here to maintain image visibility.

Across all 14 SRC images, the residuals exhibit reduced energy after restoration (see Figure 7 for a typical case; see Supplementary Figures S6 and S7 for the signed differences and forward-model residuals of all cases), while the PSNR reaches up to 31.7 dB and the SSIM up to 0.78 (Table 1), indicating improved structural fidelity and reduced noise amplification. All evaluations follow Section 2.3 settings (SNR = 16 dB; 81 px horizontal edge taper).

We further report $SSIM(\hat{F}, G)$ and $PSNR(\hat{F}, G)$ metrics to assess restored images directly against their blurred counterparts (Table 1). While images with the longest PSFs (e.g., orbit 14776) exhibit relatively low SSIM due to strong smear and noise, the farthest-distance case (orbit 7926; PSF ~11 pixels) performs markedly better. This reflects the

fundamental trade-off between sharpness recovery and noise amplification governed by the SNR parameter: higher SNRs preserve finer details but introduce more noise, whereas lower SNRs suppress noise at the cost of resolution. Our choice of SNR = 16 dB represents a practical compromise across diverse imaging conditions.

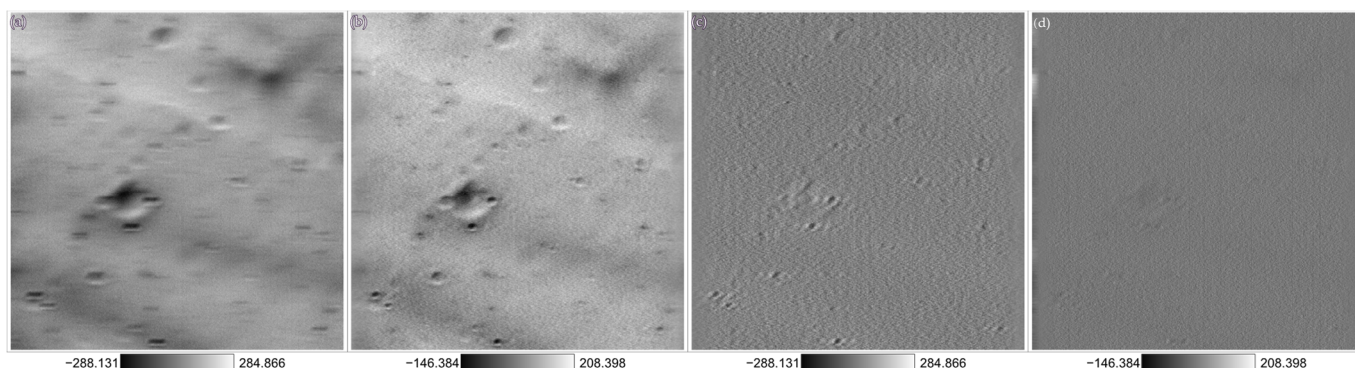


Figure 7. A representative case of validation metrics. Panels are (a) denoised input G , (b) deblurred residual $\hat{F}$, and (c) signed difference $\Delta = \hat{F} - G$, and (d) forward-model residual $R = G - (h * \hat{F})$ with $RMS(R) = 47.467$, $Vol_{gain} = 0.66$, $PSNR(h * \hat{F}, G) = 21.4515$, and $SSIM(h * \hat{F}, G) = 0.2403$. SRC image ID is h5851_0003_sr3. See Supplementary Figures S6 and S7 for signed differences and forward-model residuals on all fourteen images.

Table 1. Quantitative evaluation metrics for all 14 Mars Express SRC images after Wiener deconvolution.

Image ID	$RMS(R)$	Vol_{gain}	$SSIM(\hat{F}, G)$	$SSIM(G, h * \hat{F})$	$PSNR(\hat{F}, G)$	$PSNR(G, h * \hat{F})$
h5851_0002_sr3	55.694	0.37	0.1901	0.348	21.9605	24.5924
h5851_0003_sr3	47.467	0.66	0.0288	0.2403	17.6701	21.4515
h5851_0004_sr3	48.001	0.68	0.0675	0.3055	20.122	23.8531
h7926_0011_sr3	42.628	5.17	0.2844	0.7802	23.0024	31.6912
h8974_0002_sr3	103.986	0.43	0.3577	0.5744	24.2373	25.9505
h8974_0003_sr3	109.780	0.43	0.1627	0.3515	19.1714	20.7296
h8974_0004_sr3	121.396	0.75	0.2127	0.4251	20.0308	21.7212
h8974_0005_sr3	125.600	0.75	0.1132	0.2768	16.1278	17.6657
h8974_0006_sr3	121.874	0.71	0.2521	0.47	21.4269	22.9772
h8974_0007_sr3	119.972	0.69	0.2382	0.4499	20.8859	22.4954
h8974_0008_sr3	107.765	0.40	0.0738	0.2213	15.2278	16.7819
he776_0002_sr3	85.312	0.07	0.078	0.108	14.8854	15.5482
he776_0003_sr3	81.688	0.06	0.1432	0.1624	17.8545	18.5086
he776_0004_sr3	85.493	0.09	0.0677	0.0975	14.2231	14.8769

4. Discussion

Our motion-blur restoration pipeline—based on a priori, geometry-derived linear PSF and Wiener deconvolution—robustly recovers sub-meter Phobos surface details under diverse imaging conditions. By computing the PSF directly from spacecraft ephemeris and exposure data, we bypass the need for star-field references or manual kernel tuning, while the closed-form Wiener filter provides built-in noise regularization and avoids iteration-dependent artifacts. The extracted PSFs ranged from ~11 to ~119 pixels in length (reflecting ground-track speed and slant distance) with negligible vertical components, justifying their one-dimensional horizontal approximation.

In the high-resolution, short-exposure case (orbit 5851; 0.86 m/pixel, 14 milliseconds), deconvolution revealed crater morphologies, rim and floor textures, and linear features that were indiscernible in the original input images. Even at the coarsest resolution (orbit 7926; ~2.7 m/pixel, 14 milliseconds), grooves and crater edges sharpened with minimal

ringing, demonstrating the method's resilience to low SNR and large slant distances. For intermediate cases (orbit 8974; ~ 0.92 m/pixel, 16 milliseconds), deblurred images expose lineaments, slopes, and meter-scale boulders consistent with known Phobos morphologies. Even the most challenging dataset (orbit 14776; 0.47 m/pixel, 20 milliseconds) yielded clear views of small impact craters and positive reliefs, pushing the theoretical restoration limit under severe motion smear. For this most challenging case, a hybrid approach combining Wiener initialization followed by limited Lucy–Richardson iterations may further suppress residual artifacts. We leave systematic testing of such hybrid schemes for future work.

Compared to spatial-domain, iterative approaches like Lucy–Richardson—which have been applied to SRC and other planetary imagery—our frequency-domain Wiener deconvolution delivers a one-shot solution with explicit control over the noise–sharpness trade-off and guaranteed convergence. Because the PSF is derived from precise SPICE kernels and shape models, our framework is completely reproducible and readily adaptable to other spacecraft instruments suffering motion-induced blur.

Scientifically, recovering sub-meter features enables improved crater counting, regolith maturity analysis, and the generation of high-fidelity digital elevation models on Phobos. This approach can be extended to data from missions with similar blur challenges and will be especially valuable for the forthcoming MMX mission [14,15]. Integrating our deblurring pipeline into photogrammetric and stereophotoclinometry workflows promises significant gains in topographic precision and geological interpretation.

While our pipeline assumes a one-dimensional linear PSF dominated by along-track motion, real images may also contain cross-track jitter or non-linear blur components due to attitude fluctuations. These effects could locally broaden the PSF, potentially reducing restoration sharpness. Future work will incorporate two-dimensional, spatially varying PSFs or hybrid methods to address such deviations and improve robustness.

We also adopted a fixed SNR of 16 dB; future work should explore adaptive noise estimation to further suppress residual artifacts. Extending the model to 2D, spatially varying PSFs, and hybridizing Wiener with iterative methods may yield further improvements. Recent advances in planetary image restoration span physics-based methods, which explicitly model spacecraft motion and imaging physics (e.g., [16]), and machine learning methods using deep neural networks or GANs for super-resolution under realistic noise conditions (e.g., [17]). Our SPICE-driven Wiener deconvolution belongs to the former class, emphasizing reproducibility, physical interpretability, and independence from training datasets. Finally, automating noise-term selection based on image statistics will enhance applicability across a broader range of planetary datasets.

5. Conclusions

We have demonstrated an end-to-end pipeline for restoring motion-blurred, high-resolution SRC images of Phobos by automatically deriving a linear PSF from SPICE-based geometry and applying Wiener deconvolution with built-in noise regularization. Across fourteen images spanning four orbits, our method consistently recovers sub-meter surface features and dramatically enhances scientific usability (e.g., elucidating the origins and evolutionary history of small-scale structures and their association with spectral units [18,19]). Notably, this includes the restoration of the highest-resolution SRC images of Phobos acquired to date, which had been severely blurred. This approach is fully reproducible, requires no star-field calibration, and is readily extendable to other planetary missions facing motion-induced blur.

Beyond Mars Express, this framework can be directly applied to other planetary missions experiencing motion blur, enabling improved crater morphometry, regolith maturity studies, and topographic mapping. It is particularly relevant for the upcoming MMX

mission, where high-fidelity imaging of Phobos and Deimos will be critical for sample site selection and geological interpretation.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/rs17183256/s1>, Figure S1: An SRC image example before and after denoising processing; Figure S2: Impacts of denoising processing after restoration; Figure S3: Schematic illustration of the SRC frame onboard the MEx spacecraft in motion during SRC exposure; Figure S4: Comparison of typical images before and after edge tapering; Figure S5: Geological features identified from restored SRC images; Figure S6: Signed differences $\Delta = \hat{F} - G$; Figure S7: Forward-model residuals $R = G - (h * \hat{F})$; Table S1: The list of selected MEx SRC level 3 products; Table S2: The selected MEx SRC level 3 products with high-precision observation timings; Table S3: Results of the PSF geometries as the length and orientation of spt shifts during exposure on the SRC image plane; Table S4: Practical implications of the two deconvolution methods; Data S1: ISIS cube inputs, PSFs, and deblurred outputs. Figures S1–S7 and Tables S1–S4 are summarized in the Supplementary Information. Both the Supplementary Information and Data S1 are available on Zenodo: <https://doi.org/10.5281/zenodo.17119672>.

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Data Availability Statement: Datasets of the HRSC instrument have been downloaded from the ESA Planetary Science Archive (<https://psa.esa.int>, accessed on 18 September 2025) [20]. PDS IMG datasets of SRC level3 products were available at <https://pds-geosciences.wustl.edu/mex>, accessed on 18 September 2025). We acquired MEx SPICE kernels via ISIS data server and additional SPK from <https://archives.esac.esa.int/psa/ftp/MARS-EXPRESS/SPICE/MEX-E-M-SPICE-6-V2.0>, accessed on 18 September 2025) as well as those. Python codes (tested with Python 3.11.6) for denoising and deblurring cubes are available in R.H.’s repositories, <https://github.com/ryodohemmi/public> and <https://doi.org/10.5281/zenodo.17119672>.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ALE	Abstraction Layer for Ephemerides
DN	Digital Number
ESA	European Space Agency
MEx	Mars Express
NAIF	Navigation and Ancillary Information Facility
HRSC	High Resolution Stereo Camera
ID	Identifier
ISIS	Integrated Software for Imagers and Spectrometers
PDS	Planetary Data System

PSA	Planetary Science Archive
PSF	Point Spread Function
RDR	Reduced Data Record
SNR	Signal-to-Noise Ratio
SPICE	Spacecraft, Planet, Instrument, C-matrix, Events
SRC	Super-Resolution Channel
USGS	United States Geological Survey

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